SYLLABUS
OR 6205: Deterministic Operations Research
CRN 38674 SECTION 01

Welcome to Deterministic Operations Research! This course invites you to explore how mathematical modeling can help us make better decisions. Throughout this course, we'll keep returning to questions like: How do I translate a messy real-world situation into a precise mathematical model? What is the most efficient way to solve it? How confident should I be in this solution? Who benefits from this optimization, and who might be harmed? These questions don't have simple answers, but learning to think about them is part of becoming a thoughtful practitioner.

Term: 2026 SPRING
Start and End Dates: 01/07/2026 - 04/26/2026
Credit Hours: 4hs
Course Format: In-Person classes
Instructor Information
Priscila de Azevedo Drummond (she/her)
Email: deazevedodrummond.p@northeastern.edu
Location: Richards Hall | Room 226
Class Meeting Times: Tuesday and Friday 03:25 PM - 05:05 PM ET
Office Hours/Virtual office hours: Every Thursday from 7 pm to 8 pm via Teams (link) and Monday from 11:00 am to 12:00 pm (noon) in person at Ell Hall #301 or via Teams (link). You don't need to have a specific question to come to the office hours. If you just want an accountability partner for your work, you are also welcome.
You are welcome to contact me at any time via email or Teams. I usually check them only once a day, from Monday to Saturday, by 10 am. If I don't get to respond in 48 hours (from Monday to Saturday), feel free to send me a reminder.

During our first class, we'll review the syllabus together and I welcome your input on areas such as assignment timing, participation expectations, and how we structure collaborative work. Some elements (exam dates, core learning outcomes) are fixed, but many aspects of how we work together can be shaped by your feedback. Around week 7, I'll ask for your feedback on how the course is going and whether any adjustments would support your learning. Also, an open anonymous feedback link will always be available on Canvas (link). I review anonymous feedback weekly and will address common themes in class announcements.

If you encounter any aspect of the course - whether content, teaching methods, or classroom dynamics - that create barriers to your full participation or learning. I invite you to talk to me.

Diversity Statement

You belong here. Each of you brings valuable perspectives shaped by your unique experiences, identities, and backgrounds, including but not limited to race, ethnicity, gender identity, sexual orientation, ability status, age, socioeconomic background, nationality, and religious or cultural traditions. These diverse viewpoints are not just welcomed; they are essential to our collective learning, especially in a field dedicated to optimizing results.

Course Description

Introduces theory, computation, and application of deterministic models to represent industrial operations. Includes linear programming formulation and solution using spreadsheet and algebraic languages software; simplex, big-M, revised simplex, and dual simplex algorithms for solving linear programs; introduction to the theory of simplex, fundamental insight, duality, and sensitivity analysis; transportation, assignment, and transshipment problems; shortest path, minimum spanning tree, maximum flow, minimum cost network flow problems and project networks; and discrete-state and continuous-state dynamic programming models and applications.

Course Learning Goals and Objectives

By the end of this course, beyond gaining the skills needed to apply the techniques described in the course description, you will be able to:

1. **Formulate real-world operational problems** as mathematical optimization models - linear programs, network models, or dynamic programs - articulating the assumptions and limitations inherent in each formulation.

2. **Solve optimization problems** by selecting and applying appropriate methods based on problem structure: graphical and Simplex (primal and dual) algorithm for linear programs, specialized algorithms for network problems, and recursive approaches for dynamic programs.

3. **Analyze optimization models** using duality theory and sensitivity analysis, explaining how solutions relate to their duals and how they change when parameters change.

4. **Interpret optimization results** in context to provide actionable recommendations that account for uncertainty, changing conditions, and the needs of decision-makers.

5. **Evaluate the implications of modeling decisions**, including whose interests are represented, what is optimized, and what trade-offs are embedded in the model structure.

These aren't just academic skills; they are capabilities that will serve you throughout your career in addressing how to apply OR/IE to complex real-world challenges.

Course Methodology

This course embraces project-based learning and active learning through case studies and published problems. You'll engage with material through:

- **Collaborative discussions** where diverse viewpoints strengthen solutions.
- **Modeling** problems by hand or using Python notebooks.
- Case and problem analysis to provide **recommendations**.
- **Peer learning** through group work.
- **Reflective practice** that allows you to **revise** work after feedback for additional credit (a rubric for resubmissions will be available on Canvas).

As part of our case analysis work, each student will identify one real-world scenario that could benefit from linear programming optimization. We'll work on some of these in class, probably Week 2 and Week 10, giving you practice in problem formulation with real impact.
Mistakes are learning opportunities and are expected to happen. Questions are welcomed and encouraged.

Required Textbook: Hillier F.S. and G.J. Lieberman, *Introduction to Operations Research, 11*th Edition, McGraw Hill, 2020.

Course Activities and Assignments

This course includes the following required activities and assignments:

- **Weekly Reading and Lectures:** Weekly readings and lectures provide background knowledge, terminology, and practical examples you need to understand and correctly apply fundamental concepts in operations research. You are responsible for completing the assigned textbook readings and other materials **before class.**
- **At-home activities:** Activities and discussion-based posts should be uploaded on Canvas by the specified due date (see course schedule below). Late submissions are not accepted. You will receive your graded submission plus feedback within one week of each due date. Discussion prompts will ask you to engage critically with course material, often with supporting evidence. In some activities, peers will review and respond to each other's posts.
- **Participation in class activities:** Active learning is central to this course. Some in-class activities are ungraded practice opportunities; these give you space to try new approaches and learn from mistakes without affecting your grade. A few activities will include performance-based feedback and grades(quizzes); these will be open to revision and resubmission to improve your score.
- **Exams:** Because the concepts in this course are interrelated and built on each other, you will be assessed on your cumulative understanding at three points during the term.
- **Project:** This group activity (2-3 person teams) asks you to apply course concepts to a complex, real-world problem. The project is designed with revision built in; you'll present your work in Week 14, receive feedback from peers and the instructor, and then submit your final report in Week 15. This structure ensures you can strengthen your analysis before final evaluation.

Grading/Evaluation Standards

Please refer to the following link for guidelines (link) on the grading system:

93 – 100% A 90 – 92% A-	87 – 89.9 % B+	77 – 79.9% C+	0 – 69.9% F
	83 – 86.9% B	73 – 76.9% C	
	80 – 82.9% B-	70 – 72.9% C-	

Assignment Descriptions and Grade Breakdown

Change in grade criteria after negotiation

Component / Assignment	Planned due dates	Percentage of Grade
Online activity	After each class	10%
Activities in class	During all classes	20%
Exam 1	W5	20%
Exam 2	W10	20%
Exam 3	W16	Replace Exam 1 or 2 when criteria met*
Project: Presentation and Peer feedback	W14	10%
Project: Reports	W15	20%
Total		**100%**

Detailed assignment descriptions and rubrics will be provided at least one week before due dates. Most of the assignments, including the project instructions and expectations with examples, will be published by the second week of class.

***Exam 3 can be taken to replace grade of Exam 1 or Exam 2 in addition to the self-assessment review of Exam 1 or Exam 2. For example, if you get a low grade in Exam 1, you would have the option to make a self-assessment (rubric will be provided). That won't change the grade but will allow you to replace the Exam 1 grade with the Exam 3 grade.**

Late Submissions and Attendance Policy

Your presence enriches our practical activities. Your insights during class or at-home activities directly contribute to your classmates' understanding, and their contributions shape yours. Because of that, there are no make-up assignments. Similarly, late submissions are not accepted because many activities rely on peer learning that happens in real-time.

But we know that life happens. So, the lowest 15% of all online and in-person class activities will not be counted for the final grade (except for exams and projects). The grade-drop policy gives you flexibility for life's unexpected moments while keeping consistent expectations for everyone.
The Academic Catalog (below) defines all acceptable reasons for missing a class. If you are missing an exam or the project presentation, let me know ahead of time. I will do my best to accommodate your needs, but I must respect the CAG's (and mine) time, so be mindful of your requests. You have the responsibility of choosing what you can and cannot do and when. You have agency over your learning and how much you invest in it. And I am here to support you.

Please note the following catalogue policies: *Students have the right to a limited number of excused absences for conditions listed in the Northeastern University Attendance Requirements, including absences due to specific university-sponsored activities, religious holidays, military deployment, and jury duty.*

How We Will Work Together:

This course runs on working together. I genuinely want to hear from you throughout the semester, not just through course evaluations, but also in our everyday interactions. Your feedback helps me adjust things, so they work better for everyone.

I'm hoping we can create a classroom where people listen to each other, where we respect the time and energy everyone invests, and where we can enjoy tackling complex problems together. Yes, some of this material is challenging, but something is exciting about working through it as a group.

Here's something I want to be upfront about: I'm beginning my teaching journey with you. This means I'm learning too, not just about teaching, but from all of you. Your questions might catch me off guard sometimes, your perspectives will teach me new ways to see this material, and your feedback will help me become the teacher I want to be. Even years from now, when I'm more experienced, I hope I'll still have this same openness to learning from my students.

So, let's make this a space where we can all be curious, make mistakes, support each other, and figure out together how mathematical modeling can make a difference in the world and in our lives.

How We Will Communicate

Here is how I plan to use Canvas communication tools:

- **Announcements:** I will post announcements at the end of each week. I will also use the announcements tool to communicate any important or time sensitive updates in the course. Please turn your announcement notifications on in your Canvas setting (Links to an external site.) so you also receive announcements in email. This will help you keep on top of late-breaking developments.

- **If you have questions about content or assignments:** I have set up a discussion called "Q&A". Please post questions about content and assignments here, so your classmates can benefit from the answers. I recommend that you Subscribe (Links to an external site.) to the forum so you also receive forum messages in email.
- **Email:** You can reach me at deazevedodrummond.p@northeastern.edu through your NU email account, or you can click **Inbox** in the course menu, search for our course and locate me. See Instructions for how to use the Canvas Inbox here (Links to an external site.). You can expect a response within 24 hours on weekdays, at most 48 hours. Please check your Northeastern email daily or forward it to an email address that you do check.
- **"Off and On Office Hours":** I am available for consultations over message or through videoconferences such as Zoom or Teams. Please email me to schedule an appointment outside office hours. Please refer to the first section of the syllabus for more information.

Statement on AI Use

AI tools (like ChatGPT, Claude, etc.) can support your learning when used thoughtfully. And think about how much more rewarding it will be to **connect with your peers before connecting with the AI.** They probably can help you with more than you think, and you get the bonus of strengthening your network/friendship circle.
In this course:
Encouraged uses:

- Getting discussion points regarding mathematical concepts
- "Tutoring" your work, asking for clarifying questions or using it as self-assessment tool to help your learning process.
- Improving writing clarity
- Debugging code
- Exploring alternative formulations or questioning assumptions in your models

Not permitted:

- Submitting AI-generated work as your own
- Using AI to complete assessments
- Bypassing the learning process

Remember: The goal is to develop YOUR capabilities. AI is a tool, not a substitute for your thinking. If you are in doubt, come talk to me.

NU Academic Integrity Policy

Northeastern University is committed to the principles of intellectual honesty and integrity: the NU Academic Integrity Policy can be found on the website of the Office of Student Conduct and Conflict Resolution (OSCCR).

Just as we value diverse perspectives in our classroom, we also value the authentic contributions each of you makes to our collective learning. Academic integrity forms the foundation of meaningful scholarship and mirrors the ethical responsibilities you'll carry as engineers.

Integrity in this class means celebrating your own unique voice while properly acknowledging when you build on others' ideas; embracing collaboration by listening and sharing our thoughts; asking for help whenever in doubt; and using mistakes as opportunities for growth. In summary: work together, learn from others, and CITE, CITE, CITE!

Remember that developing your own analytical and problem-solving capabilities is essential for addressing the complex challenges we study. When you submit work that authentically represents your learning journey, you build the skills needed to work as an engineer effectively.

If you're struggling with an assignment or tempted to take shortcuts, **message me.** Together, we can find strategies to support your success while maintaining the integrity that our field and the communities we serve (or will serve) deserve.

If you're unsure whether something requires citation, please ask. I'm here to help you learn conventions, and the writing center can also help you!

Required Course Hardware and Support

- Reliable internet connection (please, make use of the university facilities whenever you need, and let me know if you need any additional support)
- Access to computer with Google Colab capability (as decided in syllabus negotiation)

Prior to taking this course, review the basic computer specifications for Canvas (Links to an external site.). You can access Canvas on the go using the Canvas Student app (Links to an external site.). Not all features work on the mobile app, I recommend completing essential course work using your computer and a web browser.
If you need help using Canvas, online chat and phone support are available 24/7 by clicking on the help menu in the left-hand navigation. For additional help, around-the-clock Northeastern tech support is still available students. For technology questions or assistance, contact the IT Service Desk 24 hours a day, 7 days a week, learn about ways to contact ITS Service Desk here (Links to an external site.).

Title IX Protections and Resources

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal funding. Any NU community member who has experienced such discrimination, sexual assault, relationship violence, stalking, coercion, and/or sexual harassment, is encouraged to seek help. Confidential support and guidance can be found through University Health and Counseling Services, the Northeastern Center for Spirituality, Dialogue, and Service, and the Office of Prevention and Education at Northeastern (OPEN). Note that faculty members are considered "responsible employees" at Northeastern University, meaning they are required to report all allegations of sex or gender-based discrimination to the Title IX Coordinator. For more information, please access Resources Overview – Office for University Equity and Compliance.

Other resources for your success:

Make use of all the university has to offer you! And if you need help navigate those services, you can ask for my help.

Disability Resource Center
The university's Disability Resource Center works with students and faculty to provide students who qualify under the Americans With Disabilities Act with accommodations that allow them to participate fully in the activities at the university. Ordinarily, students receiving such accommodations will deliver teacher notification letters at the beginning of the semester. Students have the right to choose whether to disclose their specific disabilities to instructors but must provide a letter to receive accommodations.

WeCare
WeCare offers supports for students during times of difficulty or challenge. You can find WeCare at 226 Curry Student center Monday - Friday from 8:30-5:00, call at 617-373-7591, or email wecare@northeastern.edu.

Mental Health Resources
In addition to mental health resources available through Northeastern's University Health and Counseling Services Northeastern has added Find@Northeastern, which is a 24/7 mental health consulting line and can be reached at 1-877-223-9477.

The Writing Center
The Northeastern University Writing Center offers free and friendly tutoring for any level of writer, including help with conceptualizing writing projects, the writing process, and using sources effectively. Currently, the Writing Center has virtual appointments only. To make an appointment, or learn more about the Writing Center, visit https://www.northeastern.edu/writingcenter, or email WritingCenter@northeastern.edu. Advance and same-day appointments are available.

Peer Tutoring

The Peer Tutoring Program offers a wide range of tutoring services to meet the academic needs of undergraduate students. If you need academic assistance, contact the Peer Tutoring Program Monday through Friday from 9:00am to 5:30pm. Peer tutoring services are free and open to all NU undergraduate students. Peer tutoring begins the second week of classes and ends the last day of classes. The Peer Tutoring Program is located in 1 Meserve Hall. Call 617- 373-8931, email peertutoring@northeastern.edu, or visit the weblink above.

International Tutoring Center

The International Tutoring Center (ITC) provides current Northeastern University international students with free, comprehensive English language and academic support. The ITC includes English as a Second Language Tutoring (ESL), Language and Culture Workshops, and Reading Workshops. For more information on available workshops and tutoring opportunities please visit the ITC weblink above.

Snell Library

Snell Library offers a variety of resources for undergraduate research, including subject-specific Research Guides, help with citation and bibliography, and 24/7 chat support . The library also houses the Digital Media Commons, which offers a variety of resources for instructors and students for multimedia projects.

Outreach, Engagement, Belonging: Northeastern University is committed to fostering a community of belonging, which is essential to the advancement of Northeastern University's mission of teaching and research. Our university is stronger because of the varied backgrounds, experiences, and perspectives that all members of our global community bring to the pursuit of knowledge. Embracing this pluralism is not the work of one office, department, or academic unit. It is a shared responsibility that spans disciplines and boundaries. By harnessing the power of our differences, we will continue to light the path to bold new ideas and life-changing discoveries. Please visit Belonging at Northeastern for more information.

Final statement

This syllabus is our initial agreement, but learning is dynamic, and the classes and syllabus are too. I welcome your feedback throughout the semester on how we can improve the course. Your success is my primary goal, and I'm here to support your journey in mastering the key concepts of Deterministic Operations Research.

Questions? Concerns? Ideas? Want to chat? My door is always open.

Tentative Class Schedule / Topical Outline:

Please note: for more information about specific assignments and due dates, please see instructions on your course map linked to the syllabus tab.

Module	Topic	Date	Subtopics
Mod 1	Introduction and Syllabus negotiation	9-Jan	Get to know each other. Set expectations between you (the students) and me (the instructor). Negotiation and communication skills. Self-assessment skills
	Introduction to Operations Research & Linear Programming	13-Jan	OR: why do we use it? Present an overview of OR framework. Decision Making, models, and who makes the decision. Modeling Strategies
Mod 2	Generalized LP Model and solving it by hand	16-Jan	Components of a linear program: decision variables, objective function, constraints, and their standard forms. Geometric interpretation of feasible regions, extreme points, and optimal solutions. Steps of the graphical method for two-variable LPs.
		20-Jan	Matrix form and the use of matrix operations to work through the system of equations with multiple variables.
Mod 3	Introduction to Simplex	23-Jan	Simplex method operations: pivot selection, tableau updates, optimality and feasibility conditions
		27-Jan	Simplex Method and the use of software to solve bigger problems.
	Simplex, continued: Breaking Ties and Dealing with Non-Standard Forms	30-Jan	Breaking Ties and Degeneracy + Review
		3-Feb	Big-M and two-phase methods for handling non-standard forms
	Exam #1	6-Feb	Module 1 -3
Mod 4	Theory of the Simplex Algorithm	10-Feb	Fundamental Insight
		13-Feb	Fundamental Insight and Computational complexity classifications and their implications.
Mod 5	Duality Theory	17-Feb	The relationship between primal and dual linear programs. Fundamental insight connecting primal and dual solutions
		20-Feb	Complementary slackness conditions Apply revised simplex and dual simplex algorithms Course Feedback
Mod 6	Sensitivity Analysis	24-Feb	Sensitivity analysis - shadow prices and reduced costs.
		27-Feb	Sensitivity analysis - changing in parameters and variables
	Spring Break	3-Mar	
		6-Mar	
	Exam #2	10-Mar	Module 4-6

Module	Topic	Date	Subtopics
Mod 7	Classes of network optimization problems and their relationships	13-Mar	Transportation and Transshipment Problems structure and algorithms
		17-Mar	Transportation and Transshipment Problems structure and algorithms
		20-Mar	Assignment Problem and its Algorithm
Mod 8		24-Mar	Network Optimization (shortest path, maximum flow, minimum cost flow, minimum spanning tree)
		27-Mar	
		31-Mar	Project Networks and Project Review
	Good Friday	3-Apr	
	Project Presentation and Peer Feedback	7-Apr	Project: Communicate recommendations to non-technical audiences. Defend modeling choices.
Mini	Introduction to Dynamic Programming	10-Apr	Set up and solve dynamic programs using recursive reasoning.
	Introduction to Dynamic Programming-Additional Applications	14-Apr	
	Continuous State Dynamic Programming and Review	17-Apr	Project: document assumptions. Offer recommendation of action based on model results. Evaluate implications of modeling decisions.
	Final Exam ? TBD later the specific day	21-Apr	Module 7-9
		24-Apr	